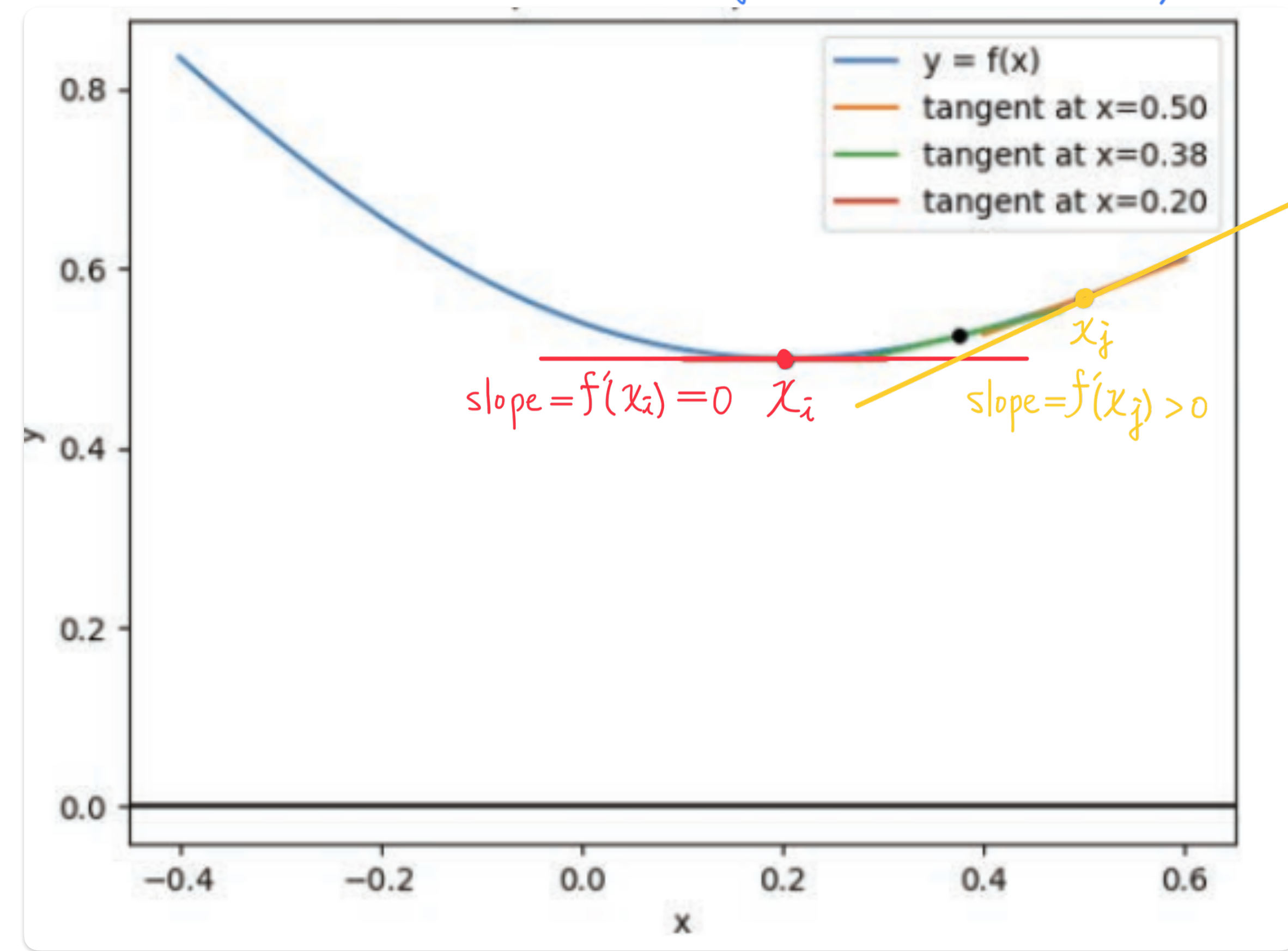


1. Concept of Derivative in "Mathematics":

A. given $y=f(x)$ function, the derivative at the point x_i ($f'(x_i)$) means the slope of tangent line which pass through the point x_i .

B. the derivative at the extreme point x_i which maximizes or minimize the value y is 0 ($f'(x_i)=0$).

C. if we move to the point (x_j) which further away from extreme point x_i , then derivative at the point x_j ($f'(x_j)$) will increase or decrease from 0.



D. given $y=f(x_0, x_1, \dots, x_n)$ function, the partial derivative of variable x_0 means pretending all variables except x_0 as constant and computing derivative of variable x_0 (the only variable that isn't treated as a constant) $\equiv \frac{\partial y}{\partial x_0}$.
the partial derivative of variable x_{n+1} means pretending all variables except x_{n+1} as constant and computing derivative of variable x_{n+1} (the only variable that isn't treated as a constant) $\equiv \frac{\partial y}{\partial x_{n+1}}$.

E. given $y=f(x_0, x_1, \dots, x_n)$ function, computing all the variables into corresponding partial derivatives, and arranging all these partial derivatives in a vector,
then we get "gradient of $y=f(x_0, x_1, \dots, x_n)$ function" $\equiv \nabla y = \begin{bmatrix} \frac{\partial y}{\partial x_0} \\ \frac{\partial y}{\partial x_1} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix}$ or "gradient descent of $y=f(x_0, x_1, \dots, x_n)$ function" $\equiv -\nabla y = -\begin{bmatrix} \frac{\partial y}{\partial x_0} \\ \frac{\partial y}{\partial x_1} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix}$.

F. Being a vector, gradient of $y=f(x_0, x_1, \dots, x_n)$ function or gradient descent of $y=f(x_0, x_1, \dots, x_n)$ function all consist of direction and magnitude.

$\Rightarrow$ direction of gradient indicates where to move from a given point ($a, b, \dots, z$) for $f(x_0, x_1, \dots, x_n)$ function can increase value of y the most,
on the other words, it is the direction of steepest ascent.

$\Rightarrow$ direction of gradient descent indicates where to move from a given point ($a, b, \dots, z$) for $f(x_0, x_1, \dots, x_n)$ function can reduce value of y the most,
on the other words, it is the direction of steepest descent.

$\Rightarrow$ magnitude of gradient indicates the slope of direction of gradient (=direction of steepest ascent).

$\Rightarrow$ magnitude of gradient descent indicates the slope of direction of gradient descent (=direction of steepest descent).